TRUNKBELAY · MARKET AND VIABILITY

TrunkBelay market and viability

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First market and viability note (kit 1.9.1 upgrade)

TrunkBelay market and viability

TrunkBelay is for coconut climbers who rescue each other, the training schemes that teach them and the rural fire crews who arrive later; it competes with improvised cloth and rope and, at a distance, with utility pole-top rescue kits that cost about USD 1,576 and are not made for palms. An open design matters because no one sells a palm rescue anchor, and because most of the kit's cost is certified rescue gear that a training scheme can buy in bulk. Prices and benchmarks are cited with the date they were checked. Method: STANDARDS section 23.

> **Safety:** TrunkBelay is not certified equipment and is not sold. Any route to wider use runs through the TRL 4 proof test, the dummy drill and a partner's training programme.

Table 1. Market segments.

Segment	Target users and region	Competitive benchmark (cited price)	Estimated cost per unit (USD)	Value proposition
Peer rescue by coconut climbers	Climber groups and labour banks in Kerala and Tamil Nadu first	Improvised cloth and rope (no cost, no anchor); fire crews with ladders and nets (TKB-PRB-001)	828 per kit (BOM, single-kit prices); one kit per climber group	A rescue in minutes instead of a wait of 30 to 150 minutes head down
Climber training schemes	Friends of Coconut Tree training under the Coconut Development Board	No rescue kit or drill in current training (TKB-PRB-001)	828, less group purchase of the rescue gear (not counted)	A standard kit and drill to teach in half a day (PR-010)
Fire and rescue services	Rural stations with palm rescues (Kasaragod, Kannur)	Utility pole-top rescue kit: Buckingham SuperSqueeze 488P, about USD 1,576, rated 350 lb, for smooth poles (as cited in TKB-PRB-001, checked 2026-09-30)	828	A light trunk anchor to secure a victim before ladders or nets are set
Bought rescue gear inside the kit	Any of the above	Auto-locking descender: Petzl I'D S, USD 421.20 (Haness Supply, checked 2026-10-25); Climbing Technology Sparrow 200R, GBP 179.95 (Skyland Equipment, checked 2026-10-25). Evacuation triangle: Petzl Pitagor, USD 92.90 on offer (Oliunid, checked 2026-10-25)	BOM carries USD 280 for the descender class and USD 140 for the triangle class	Specification by standard, not brand, lets a partner buy the cheapest certified device
Other single-stem palms	Toddy tappers, arecanut and date palm workers	None found	828	Same collar on trunks within 200 to 450 mm

The BOM's USD 280 for the descender sits between the two cited devices; a branded I'D S would add about USD 141 and take the kit to about USD 969, over PR-009's USD 850. The triangle line (USD 140) is above the one cited offer; a cheaper certified triangle would take about USD 47 off. Neither is changed in the BOM: both are class prices from the TRL 3 session, to be replaced by quotes from the partner's supplier (TKB-DEC-001, item 3).

Open route

- **Who builds it.** A fabrication shop with an AC TIG welder makes the frame (about USD 69 of the BOM, TIG welding included); everything else is bought to specification. A training scheme or a climbers' collective would hold the kits, one per group.
- **How it spreads.** Through the Friends of Coconut Tree training programme and fire service stations that have done palm rescues: a drill card, the build plan (TKB-BLD-001) and the open drawings let other districts and other palm crops copy it.
- **What keeps it open.** CERN-OHL-S-2.0 for the hardware and MIT for the scripts; the design keeps clear of the Cynch-Lok ratcheting cinch (PR-017); published with a DOI on Zenodo.
- **What it is not.** Not a product for sale and not certified equipment; any partner who wants a certified version faces EN 795-class testing at 12 kN (TKB-REG-001, G2).

Sources

- Haness Supply. "Petzl D020AA00 I'D S Descender/Belay Device." https://www.hanessupply.com/petzl-d020aa00 (checked 2026-10-25).
- Skyland Equipment. "CT Sparrow 200 R Rescue Descender." https://skylandequipment.com/products/ct-sparrow-200 (checked 2026-10-25).
- Oliunid. "Petzl Pitagor evacuation triangle." https://www.oliunid.com/petzl-pitagor-evacuation-triangle (checked 2026-10-25).
- Buckingham Manufacturing. "SuperSqueeze Pole Top Rescue 488P." https://buckinghammfg.com/products/supersqueeze-pole-top-rescue-488p/ (as cited in TKB-PRB-001, checked 2026-09-30; not rechecked).
- TKB-PRB-001 for the reported rescues and the training scheme.